



Project Proposal

Thesis Title: Concise and Engaging Title

by

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1 Introduction

By training for next-word prediction in large datasets of texts, language models learn to encode linguistic properties that can be leveraged in a wide variety of downstream language-related tasks. In particular, transformer-based language models have become a cornerstone technology in Natural Language Processing (NLP) by advancing the state-of-the-art in many of its tasks. However, their data-hungry nature limits their applicability to the few languages that have sufficiently large datasets for language models to train on.

One solution to this limitation is to have language models train on texts written in a variety of languages. By doing so, such models pick up on near-universal linguistic properties, such as there being a distinction between nouns and verbs [7]; as well as properties shared only by subsets of languages, such as verbs preceding their subject in a sentence. This information is then encoded in a shared semantic space that allows multilingual language-models to use information obtained from one language when processing another in a process called cross-lingual transfer. This is particularly useful in the low-resource setting: a language model may not have trained on enough data in Scottish Gaelic to have captured that verbs precede their subject in the language. However, if the model has been trained on enough texts in Modern Standard Arabic, it may well have managed to encode that verbs precede their subjects in that language, and consequently transfer that knowledge cross-lingually to better deal with texts in Scottish Gaelic.

In a process called zero-shot transfer, Pires et al. [15] finetune the multilingual language model M-BERT [?] on a source language and evaluate it on a different target language. They report remarkably high performance on the target languages, indicating that multilingual models can be the basis of successful classification models even when no training data is available in a given language. Lauscher et al. [10] report considerable improvements over zero-shot transfer when first finetuning on as few as 10 examples in a target language. Moreover, they find M-BERT's cross-lingual generalizability to be closely linked to how well-represented a target language is in M-BERT's pre-training data. Both Lauscher et al. [10] and Pires et al. [15] report cross-lingual transfer to be more successful when the source and target language are typologically similar.

Linguistic typology is the study of which linguistic properties are exhibited by which languages [4]. Choenni & Shutova [2] investigate how typology is encoded in the embedding space of – among others – M-BERT. They show that the model carves its embedding space into language-specific subspaces during pre-training. That is, sentence embeddings in a given language tend to cluster around a language-specific centroid. Moreover, probing experiments reveal that typological properties are encoded jointly across languages. For example, subtracting the Modern Standard Arabic centroid from

a Scottish Gaelic sentence embedding would remove the information from the embedding that verbs precede their subject. This indicates overlapping subspaces for typologically similar languages, which could be what underlies the relation between typological similarity and successful cross-lingual transfer.

The typologically-informed overlap between language-specific subspaces in a language model’s embedding space could be what underlies cross-lingual transfer, especially since cross-lingual transfer seems to be more successful for typologically similar languages. The goal of this project is to put that intuition to scrutiny. This is done in a three-stage setup where the model is consecutively finetuned to a high-resource source language, one of various auxiliary languages, and finally for a few examples to one of various low-resource target languages. The specific aim is then to answer the following questions:

- Q1:** Is there a positive correlation between the typological similarity and cross-lingual transfer performance?
- Q2:** Is few-shot transfer from a source to a target language more effective when first fine-tuning on a typologically similar auxiliary language?
- Q3:** Does fine-tuning on a language make its typological features more easily identifiable in the model’s embedding space?

2 Related Work

Multilingual BERT (M-BERT) [?] is a transformer-based language model that has been fine-tuned on monolingual Wikipedia corpora from 104 languages, learning to map text from each of those languages to a shared embedding space. Pires et al. [15] investigated its cross-lingual generalizability by doing zero-shot transfer. Quantifying typological similarity as the number of shared word-order related features, they find typological similarity to be positively correlated with zero-shot performance. Lauscher et al. [10] similarly find zero-shot transfer to be most effective across typologically similar languages, but find that few-shot transfer – training on as few as 10 examples in the target language – yields considerable improvement over zero-shot transfer.

Further improvements in a low-resource setting can be obtained by adding an intermediary step with auxiliary languages. Nooralahzadeh et al. [14] use meta-learning as an intermediary step for improved performance on the tasks of natural language inference (NLI) and question answering (QA). In their setup, they first finetune M-BERT on a large, English-language dataset for the task at hand. Next, they use meta-learning to optimize the model parameters for quick adaptation to new languages. In short, it involves taking model parameters θ from the first stage, fine-tuning on an auxiliary language i to obtain θ_i , and taking the derivative of the loss $\mathcal{L}(\theta_i)$ with respect to the original parameters θ to update θ with. The result is a set of parameters optimized for adaptation to a new language. While they find this setup leads to improved zero-shot performance overall, they find that languages which share certain typological features benefit from this setup most.

Langedijk et al. [9] adapt this setup by using a typologically diverse set of auxiliary languages and focus on the task of dependency parsing. They find improvements over

a zero-shot baseline are largest for target languages that are typologically distant from the source language, possibly because the auxiliary languages provide the model with new linguistic information that can later be leveraged. Choenni et al. [1] add language-specific subnetworks to the meta-learning setup. That is, they identify which subsets of parameters are most used when dealing with a given language, and restrict the model to only use that subset when processing that language. Similarly, they find that this setup yields greater improvements for typologically dissimilar source-target pairs. However, a setup that uses subnetworks without using meta-learning found the opposite effect: better improvements for more similar source and target languages.

Merchant et al. [12] investigate how the information encoded by BERT shifts during finetuning. In particular, they train probes to identify various linguistic features from a learned scalar mixing of the representations from each of BERT’s layers. They then compare the performance of these probes on a pre-trained BERT with the performance on a BERT-base model finetuned for NLI, QA, and dependency parsing. While finetuning typically does not alter the presence of these features BERT’s embeddings much, finetuning for dependency parsing causes syntax-related features to be more clearly identifiable.

Meanwhile, as stated, Choenni & Shutova [2] show that M-BERT carves its embedding space into language-specific subspaces, with typological features encoded jointly across languages. Combining this with Merchant et al.’s [12] findings, it is conceivable that finetuning on a specific language amplifies typological features relevant to that language in the model’s parameters. In turn, these features then become more readily available to any target language that shares them, which could explain the effectiveness of multistage cross-lingual transfer.

3 Methodology

The methodology involves a three-stage finetuning setup outlined in Section 3.2, with the model being fine-tuned for dependency parsing in a variety of languages. The choice for dependency parsing is based on the availability of uniform datasets in a wide variety of languages through Universal Dependencies (UD) [13]. The selection of languages is clarified in Section 3.3, the specifics of the model are discussed in Section 3.1.

3.1 Model

The model used is derived from UDify, [8] which is itself based on M-BERT [5]. UDify incorporates a layer attention mechanism that scales each layer for its importance to the task at hand. That is, during fine-tuning, a weighted sum $\mathbf{r}_j$ is computed for input token j over all layers $i \in [1, \dots, 12]$ as follows:

$$\mathbf{r}_j = \eta \sum_i \mathbf{U}_{i,j} \cdot \text{softmax}(\lambda)_i$$

where $\mathbf{U}_{i,j}$ is the output of layer i at token position j , η a trainable scalar, and λ a vector of trainable scalar mixing weights. Next, $\mathbf{r}_j$ is passed through two separate biaffine attention classifiers [6] for predicting the arc and the tag, respectively. Finally, the Chu-Liu/Egmonds algorithm [3] is used to obtain a valid dependency tree.

3.2 Fine-tuning setup

As stated, the fine-tuning setup follows three stages. In the first stage, the model M is fine-tuned on a German-language (*deu*) treebank to instill general knowledge of the task at hand. In the second stage, four copies of M are fine-tuned separately on treebanks of auxiliary languages $\ell \in \{nld, norces, fin\}$ ¹ to obtain four instances of M^ℓ . Finally, each M^ℓ is fine-tuned on each target language $\lambda \in \{gsw, fao, hsb, vep\}$,² resulting in 16 instances of $M^{\ell, \lambda}$. If time permits, both the second and third stage are ideally repeated across at least five random seeds.

3.3 Languages

Broadly speaking, language selection is based on their typological similarity and their prevalence in M-BERT’s pre-training data as reported by Wu & Dredze. [18] Typological similarity is defined using the syntactic feature-vectors (`syntax_knn`) from the URIEL knowledge base. [11] In particular, each language is represented as a feature-vector, and similarity between languages is taken to be the cosine similarity between their respective vectors.

German was selected as one of the most prominent languages in M-BERT’s pre-training data. Moreover, Turc et al. [17] found German (alongside Russian) to be the most effective language for cross-lingual transfer. The selection of the auxiliary languages is motivated phylogenetically, as they each represent a different level of relatedness to German (Finnish is unrelated, Czech is in the same family, Norwegian is in the same branch, Dutch is in the same subbranch). However, a hierarchical clustering analysis of the `syntax_knn` vectors (Appendix A) found their selection to be typologically valid, too. The target languages were selected as being the most typologically similar language to each of the auxiliary languages that (a) have a UD treebank and (b) are not part of M-BERT’s pretraining data.

4 Experimental Setup

The goal of the project is twofold: Firstly, it aims to show how typology relates to model parameter updates in the fine-tuning stage; secondly, it aims to investigate the effect of typological distance on the success of cross-lingual transfer. Section 4.1 shows how the setup outlined in Section 3.2 serves to achieve these goals. Section 4.2 lists the parameters used, Table B in Appendix B lists the specific datasets used in the experiments.

4.1 Analyses

Q1 In the second stage, four model instances optimized for dependency parsing in German is fine-tuned to do dependency parsing in languages with varying levels of similarity to German. Intuitively, the German-language data in the preceding stage would have

¹ Dutch, Norwegian, Czech, and Finnish.

² Alemannic, Faroese, Upper Sorbian, and Veps.

prepared it for dependency parsing in the closely related Dutch more than in the unrelated Hungarian. H1 would suggest that the model instance that fine-tunes to Dutch sees less extensive parameter updates than the one that fine-tunes to Hungarian. In addition, H2 suggests that the updates should primarily be to the earlier layers of the model for Dutch, whereas for Hungarian, they would be across all layers.

The fine-tuning setup includes learned scalar weights $\mathbf{w}_i$ that indicate the importance of each layer i . Following Tenney et al. [16], these can be used to detect where model updates took place for each ℓ . With $\Delta\mathbf{w}^\ell = \mathbf{w}^\ell - \mathbf{w}^{deu}$ being the shift in layer importance after fine-tuning on ℓ , the expectation is that $\Delta\mathbf{w}_i^{nld}$ is greater for layers i that deal with lexical information (i.e., the earlier ones [16]), but lower for layers i that deal with higher-level typological features, since those are generally shared between Dutch and German. On the other hand, the values of $\Delta\mathbf{w}^{hum}$ are expected to be more uniform, since Hungarian differs significantly from German both lexically and syntactically.

Q2 In the third stage, the model instances for languages ℓ are fine-tuned further to each the low-resource languages λ . Each $M^{\ell,\lambda}$ is then evaluated on the test set for λ , with performance measured as the Labeled Attachment Score (LAS). H3 suggests that greater cosine similarity $s(\ell, \lambda)$ should correspond to higher performance on λ 's test set.

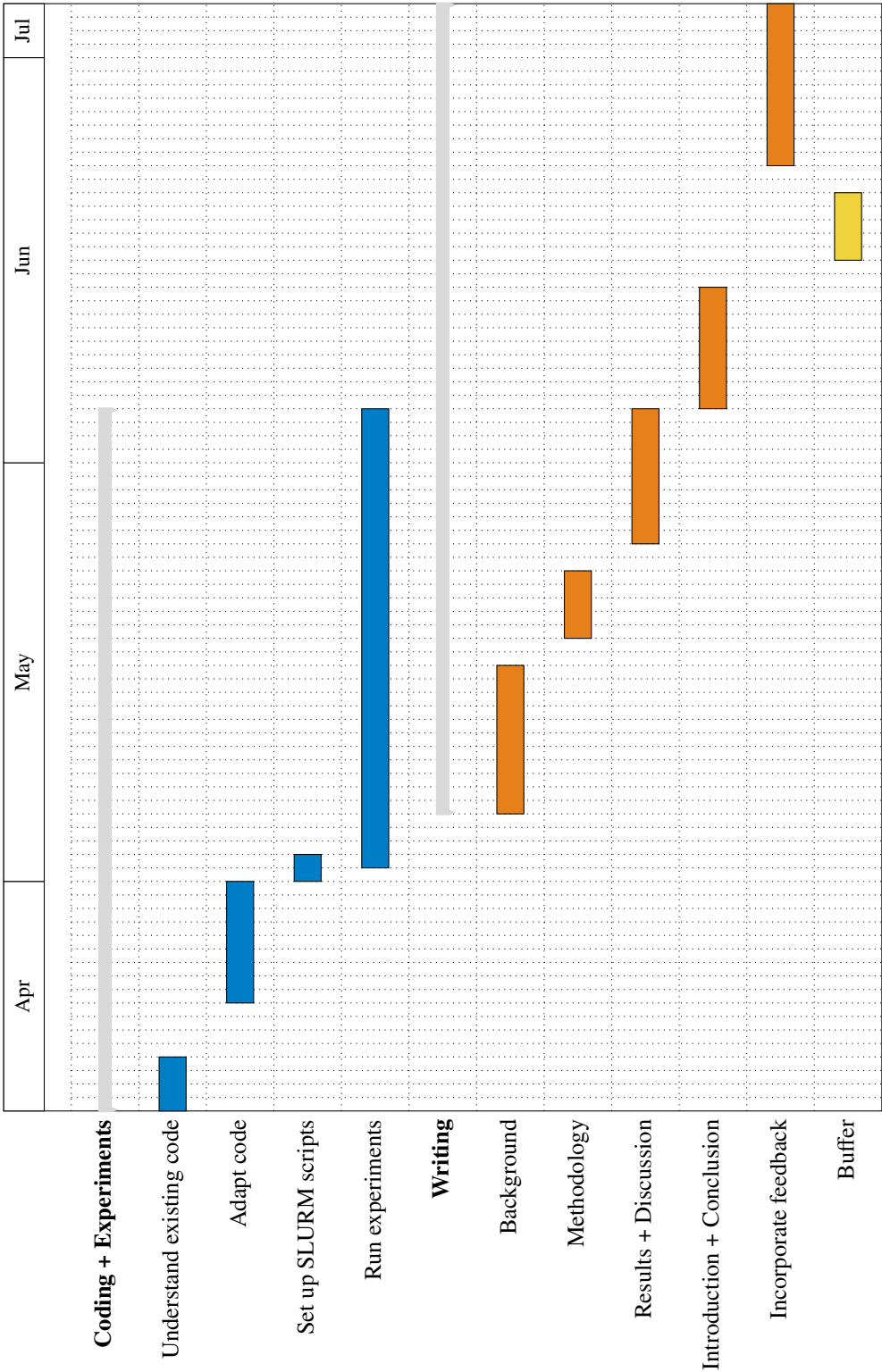
4.2 Parameters

The model is first fine-tuned to Part A³ of the German training set. Choenni et al. [1] use the English EWT treebank with 12.5k samples in this stage and train for 60 epochs. To match the number of examples in this setup, training on the German treebank spans 11 epochs. In the second stage, the model instances are fine-tuned over 1 000 iterations with a batch size of 20 for all languages ℓ . In the third stage, the models are fine-tuned on 20 examples sampled randomly from each treebank of language λ . For the Upper Sorbian treebank, these are sampled from the train set. The other three treebanks do not split their data into separate sets; therefore, the sampled examples are removed from the dataset so as to prevent evaluating on them.

Following Choenni et al. [1], a cosine-based learning rate scheduler with 10 % warm-up and the Adam optimizer are used. For the language model, a learning rate of $1e - 04$ is used, and for the classifier, a learning rate of $1e - 03$ is used.

³ The HDT treebank training data is split into various parts. With Part A being larger than the entire English-language treebank used by Choenni et al. [1], the decision was made to use just that part for training.

5 Planning



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A Appendix

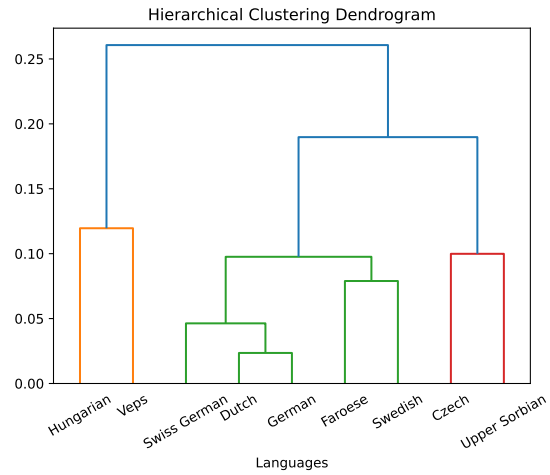


Fig. 1. Agglomerative Hierarchical Clustering dendrogram based on the cosine similarity of the `syntax_knn` vectors of the included languages.

B Appendix

Language	Treebank	Number of tokens	Number of sentences
Czech	CAC	494 142	24 709
Dutch	LassySmall	297 486	17 120
Faroese	OFT	10 002	1 208
German	HDT	3 399 390	189 928
Hungarian	Szeged	42 032	1 800
Swedish	Talbanken	96 859	6 026
Swiss German	ATB	652	98
Upper Sorbian	UFAL	11 196	646
Veps	VWT	1 303	103

Table 1. The specific UD treebanks featured in the project.